
생수다 BABP 4주차
“환경에 대한 적응의 결과, 진화”

“Collect”

실험 디자인

“Process”

필터링 & 정렬

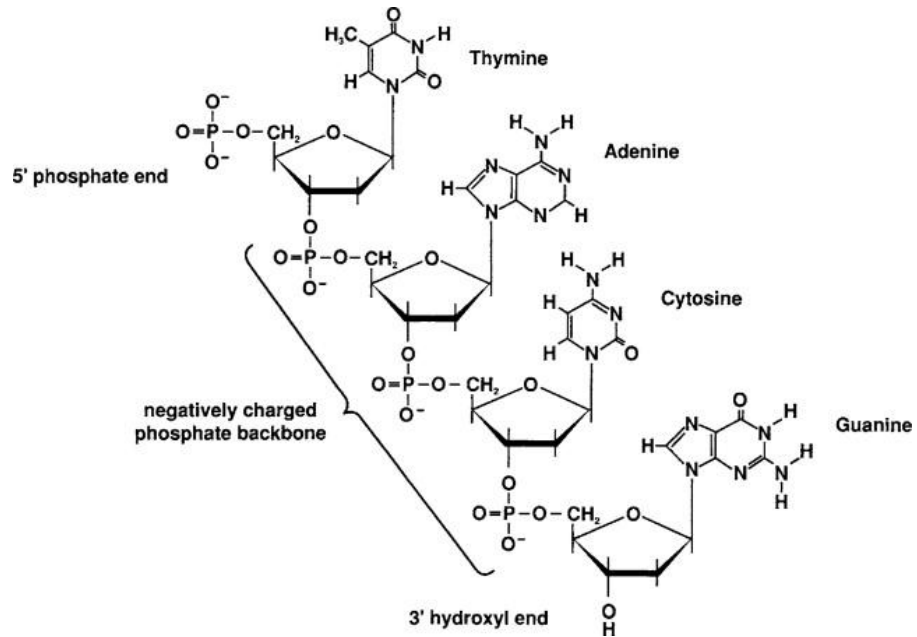
“Analyze”

시각화 & 해석

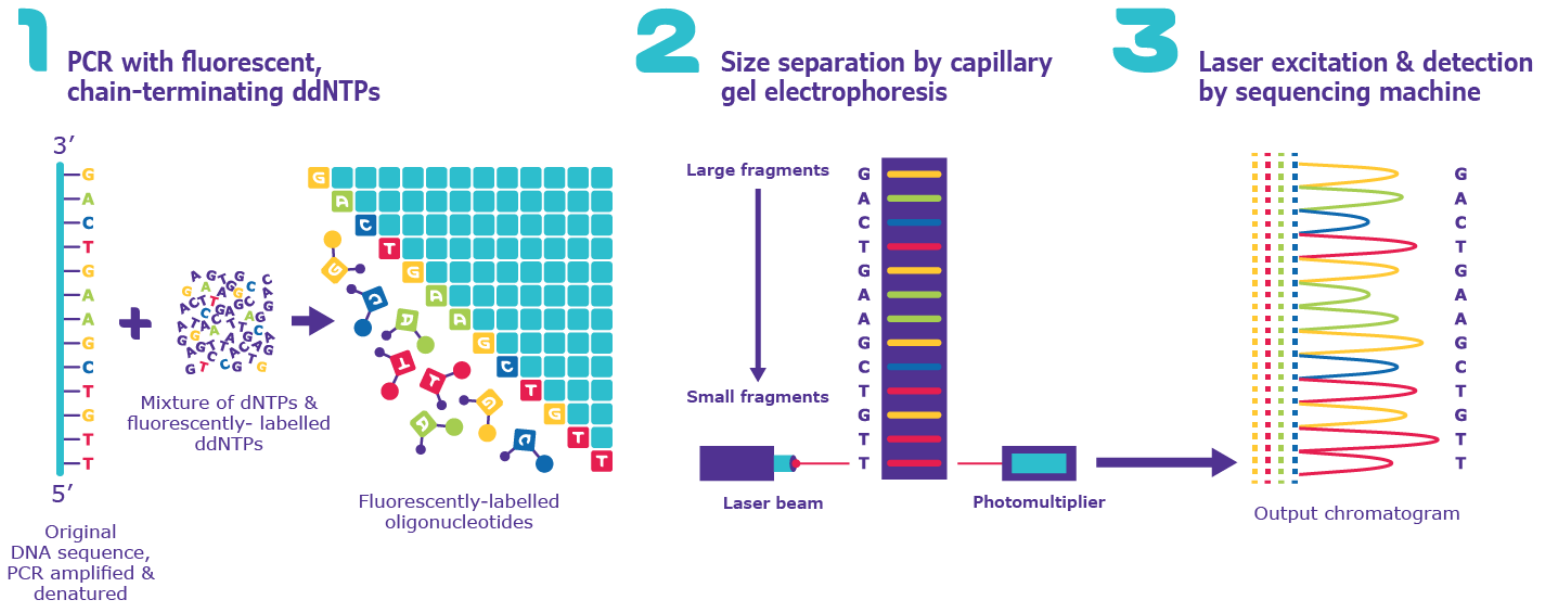
DNA Sequencing



Sanger sequencing



ScienceDirect



Sigma-Aldrich

Database for Bioinformatics: NCBI



 An official website of the United States government [Here's how you know](#) 

 **National Library of Medicine**
National Center for Biotechnology Information

Log in

All Databases 

Search

NCBI Home

Resource List (A-Z)

All Resources

Chemicals & Bioassays

Data & Software

DNA & RNA

Domains & Structures

Genes & Expression

Genetics & Medicine

Genomes & Maps

Homology

Literature

Proteins

Sequence Analysis

Taxonomy

Training & Tutorials

Variation

Welcome to NCBI

The National Center for Biotechnology Information advances science and health by providing access to biomedical and genomic information.

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Download

Transfer NCBI data to your computer



Learn

Find help documents, attend a class or watch a tutorial



Develop

Use NCBI APIs and code libraries to build applications



Analyze

Identify an NCBI tool for your data analysis task



Research

Explore NCBI research and collaborative projects



Popular Resources

- [PubMed](#)
- [Bookshelf](#)
- [PubMed Central](#)
- [BLAST](#)
- [Nucleotide](#)
- [Genome](#)
- [SNP](#)
- [Gene](#)
- [Protein](#)
- [PubChem](#)

NCBI News & Blog

Updates to the NIH Manuscript Submission (NIHMS) System 05 May 2026

In Support of the 2024 NIH Public Access Policy As part of continued

Reminder! NCBI is Retiring rsync for FTP Downloads June 1, 2026 29 Apr 2026

Announcing brownout periods when rsync will be disabled As we previously

GenBank Release 271.0 is Available! 27 Apr 2026

GenBank release 271.0 (4/14/2026) is now available on the NCBI FTP site. This release has 53.90 trillion bases and 6.27

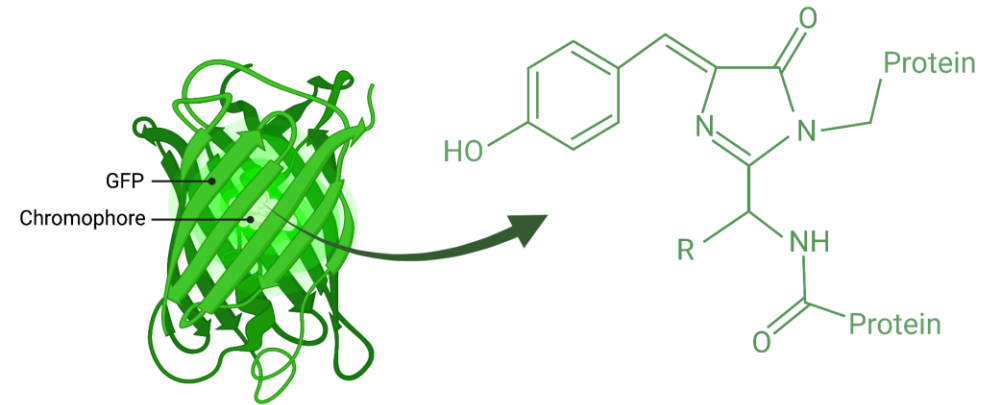
[More...](#)

Green Fluorescence Protein



FineArtAmerica

Green Fluorescence Protein (GFP) with Chromophore



BioRender

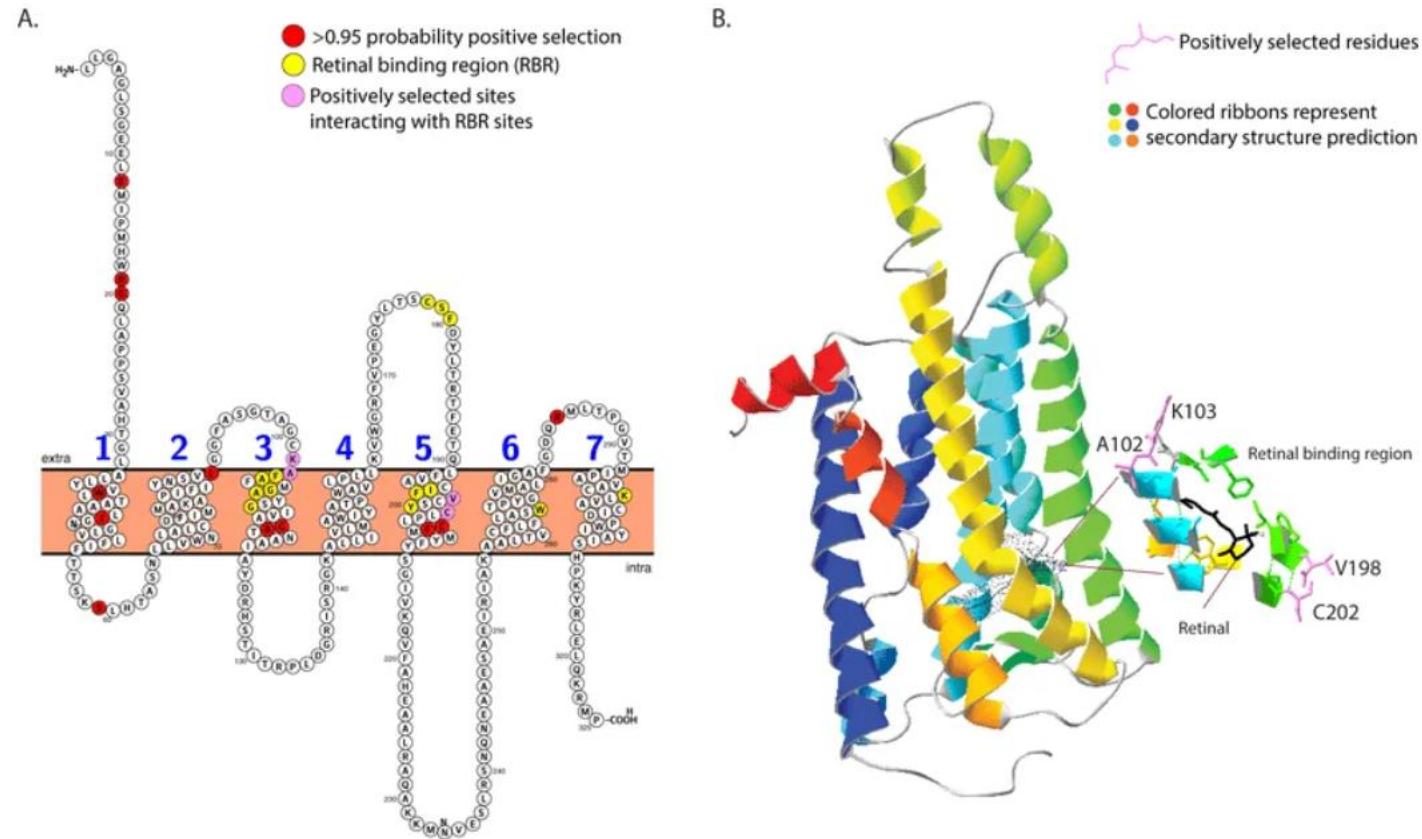
“ β -barrel”

How We Distinguish Colors



<https://www.youtube.com/watch?v=LPpzfyecrEw>

The Opisins



Sondhi *et al.* (2021).

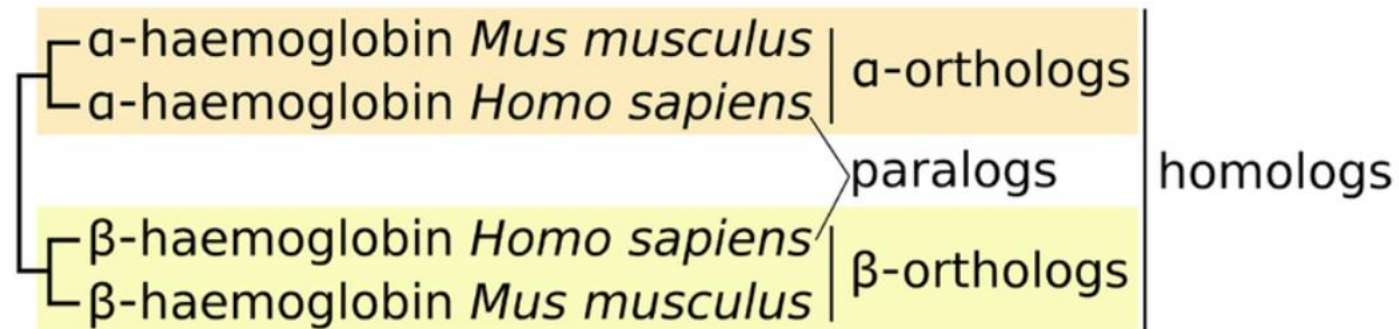
Order of Data Analysis



하나의 공통 조상으로부터 진화한 생물은 종이 분기되어도 여전히 몇몇 서열을 보존하고 있다.

Homolog: 진화 과정에서 유효한 관계가 있는 유전자 혹은 단백질

- Ortholog: 같은 기능을 가진 유전자가 여러 종으로 분지되어 생겨난 관계
- Paralog: 특정 기능을 가진 유전자가 duplication이 일어나 생겨난 관계
- + Analog: 유사한 기능을 하지만, 진화적으로 관계가 없는 유전자 혹은 단백질

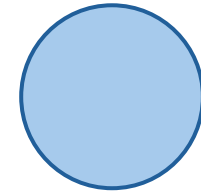
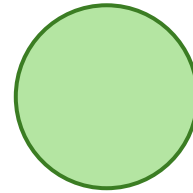
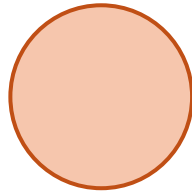


Wideman *et al.* (2018).

Opsins to Analyze



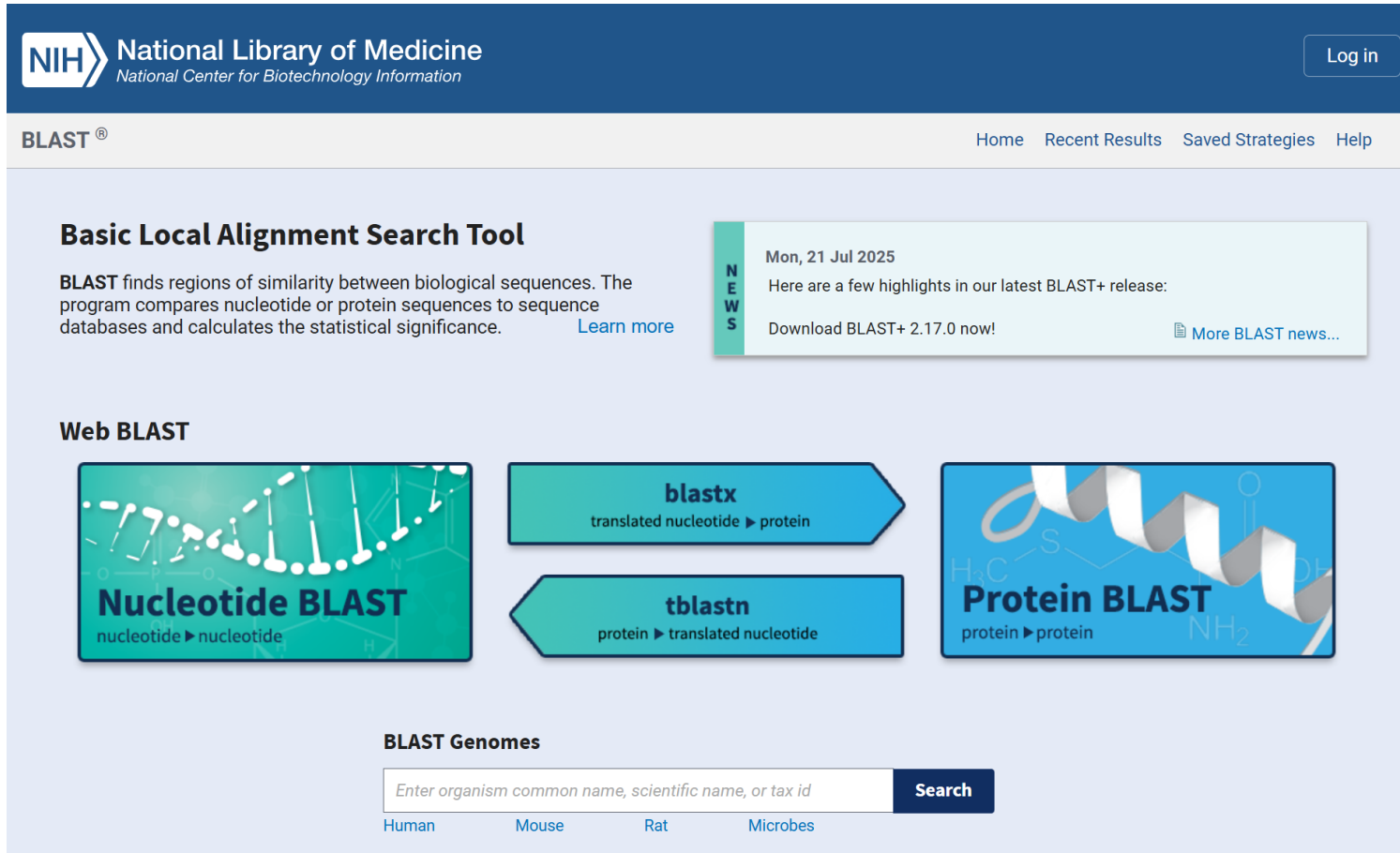
Paralog



Ortholog



BLAST: Basic Local Alignment Search Tool



The screenshot shows the NCBI BLAST homepage. At the top is the NIH National Library of Medicine header with a 'Log in' button. Below this is a navigation bar with 'BLAST®' and links for 'Home', 'Recent Results', 'Saved Strategies', and 'Help'. The main content area features a 'Basic Local Alignment Search Tool' section with a description of BLAST's function and a 'Learn more' link. To the right is a 'NEWS' sidebar with a date 'Mon, 21 Jul 2025' and a link to 'More BLAST news...'. Below the main section is the 'Web BLAST' area, which includes three large buttons: 'Nucleotide BLAST' (nucleotide to nucleotide), 'blastx' (translated nucleotide to protein), and 'tblastn' (protein to translated nucleotide). To the right of these is a 'Protein BLAST' button (protein to protein). At the bottom is the 'BLAST Genomes' section, which has a search input field with placeholder text 'Enter organism common name, scientific name, or tax id' and a 'Search' button. Below the input field are links for 'Human', 'Mouse', 'Rat', and 'Microbes'.

NIH National Library of Medicine
National Center for Biotechnology Information

Log in

BLAST® Home Recent Results Saved Strategies Help

Basic Local Alignment Search Tool

BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance. [Learn more](#)

NEWS
Mon, 21 Jul 2025
Here are a few highlights in our latest BLAST+ release:
Download BLAST+ 2.17.0 now! [More BLAST news...](#)

Web BLAST

Nucleotide BLAST
nucleotide ► nucleotide

blastx
translated nucleotide ► protein

tblastn
protein ► translated nucleotide

Protein BLAST
protein ► protein

BLAST Genomes

Enter organism common name, scientific name, or tax id **Search**

[Human](#) [Mouse](#) [Rat](#) [Microbes](#)

⇒ BLAST는 데이터베이스와의 비교 & 한 쌍의 서열을 비교하는 데에 특화

1 **blastn** blastp blastx tblastn tblastx

Align Sequences Nucleotide BLAST

BLASTN programs search nucleotide subjects using a nucleotide query. [more...](#)

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#)

Query subrange [?](#)

From

To

Or, upload file [?](#) 4 파일 선택 opn1mw_gene.fna

Job Title

Enter a descriptive title for your BLAST search [?](#)

2 ☒ Align two or more sequences [?](#)

Enter Subject Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#)

Subject subrange [?](#)

From

To

Or, upload file [?](#) 3 파일 선택 opn1lw_gene.fna

Program Selection

Optimize for

5 ☐ Highly similar sequences (megablast)

☐ More dissimilar sequences (discontiguous megablast)

☒ Somewhat similar sequences (blastn)

Choose a BLAST algorithm [?](#)

BLAST

Search nucleotide sequence using Blastn (Optimize for somewhat similar sequences)

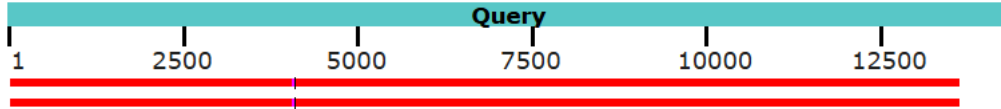
☐ Show results in a new window

- ① 한 쌍의 유전자 서열을 비교할 것이므로 ‘blastn’ 사용
- ② ‘Align two or more sequences’ 설정을 체크
- ③ 파일 선택: Subject Sequence에는 적색 옅신 유전자를 입력
- ④ 파일 선택: Query Sequence에는 각각 녹색 / 청색 옅신 유전자를 입력하여 총 두 번 수행
- ⑤ 알고리즘 ‘blastn’ 사용

BLAST Results of Opsin Paralog



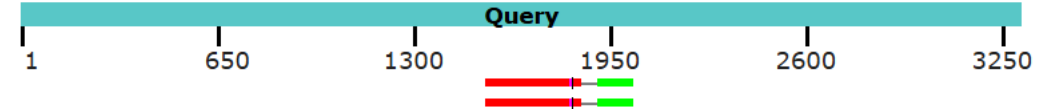
Distribution of the top 228 Blast Hits on 2 subject sequences



Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
17146	36002	95%	0.0	98.90%	14790	Query_6960992
17133	35988	95%	0.0	98.87%	14790	Query_6960993

적색 / 녹색

Distribution of the top 28 Blast Hits on 2 subject sequences



Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
239	1806	13%	9e-65	79.58%	14790	Query_8140213
239	1806	13%	9e-65	79.58%	14790	Query_8140212

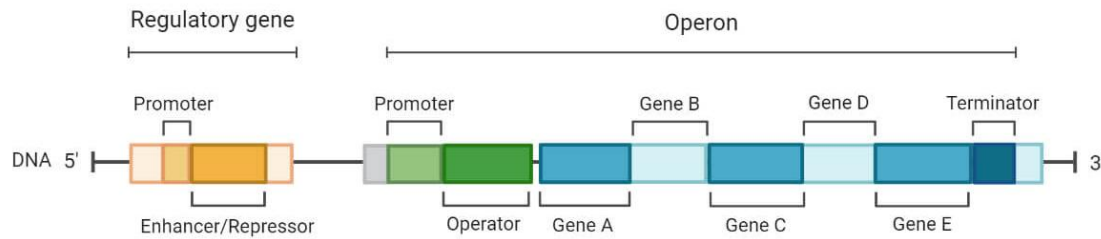
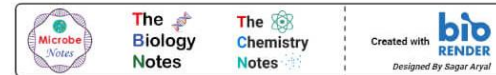
적색 / 청색

*E value: 특정 크기의 데이터베이스를 검색했을 때, 해당 Hit와 같거나 더 높은 점수의 결과가 우연히 나타날 것으로 기대되는 횟수. 값이 작을수록 우연히 발견되었을 가능성이 낮음.

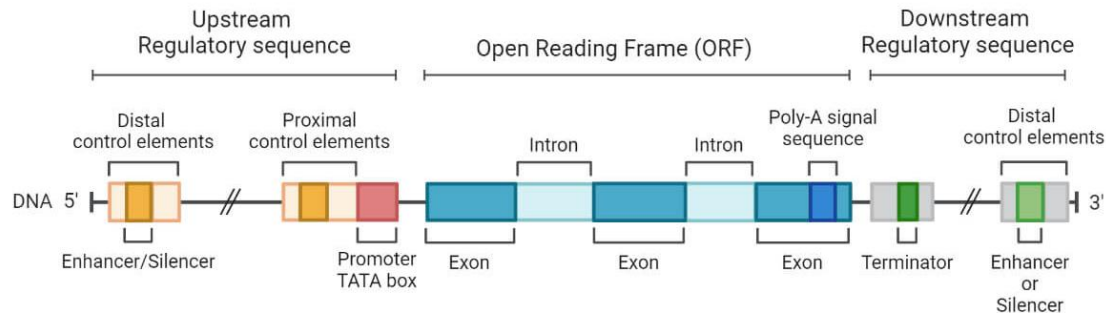
BLAST Results of Opsin Paralog



Prokaryotic Gene Structure



Eukaryotic Gene Structure



Words (Protein)

Setup

Query: GTQITVEDLFYNIATRRKALKN

GTQ

TQI

QIT

ITV

TVE

VED

EDL

DLF

...

Neighborhood Words

→ LTV, MTV, ISV, LSV, etc.

Words (Nucleotide)

Setup

Query: GTACTGGACATGGACCCTACAGGAA

11-mer

GTACTGGACAT

TACTGGACATG

ACTGGACATGG

CTGGACATGGA

TGGACATGGAC

GGACATGGACC

GACATGGACCC

ACATGGACCCT

...

Lookup table

MSA(Multiple Sequence Alignment)



Multiple Sequence Alignment

				115					120					125					
Sequence A	F	S	T	A	A	F	R	F	G	H	A	V	H	P	L	V	R	R	L
Sequence B	F	S	T	A	A	F	R	F	G	H	A	V	H	P	L	V	R	R	L
Sequence C	F	S	T	A	A	F	R	F	G	H	A	V	H	P	L	V	R	R	L
Sequence D	F	S	T	A	A	F	R	F	G	H	A	I	H	P	L	V	R	R	L
Sequence E	F	A	T	A	A	F	R	F	G	H	A	V	Q	P	I	V	R	R	L
Sequence F	F	T	T	A	A	F	R	F	G	H	A	I	P	P	M	V	H	R	L
Consensus	F	s	T	A	A	F	R	F	G	H	A	v	h	P	I	V	r	R	L

BioRender

- 길이가 비슷한 서로 다른 세 개 이상의 생물학적 서열 (핵산, 단백질)을 정렬하는 것.
- 상동성 추론 및 서열 간의 진화적 관계의 연구에 활용됨
- 오늘 사용할 알고리즘: “Clustal Omega”

